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A comparative study of the effects of gestodene 60 μg/ethinylestradiol 15 μg and desogestrel 150 μg/ethinylestradiol 20 μg on hemostatic balance, blood lipid levels and carbohydrate metabolism

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ABSTRACT

Objectives This multicenter, randomized, parallel-group, open-label study was conducted to compare the effects of gestodene (GTD) 60 μg/ethinylestradiol (EE) 15 μg and desogestrel (DSG) 150 μg/EE 20 μg (Mercilon®) on selected metabolic measurements during six cycles in 124 healthy women.

Methods GTD/EE subjects received a single pill once daily from days 1 to 24 of a 28-day cycle, followed by placebo pills daily for the last 4 days of the cycle. DSG/EE subjects received a single pill daily from days 1 to 21 of a 28-day cycle, followed by a 7-day pill-free interval. Safety was assessed from changes in hemostatic measurements, lipid profile, glucose tolerance and adverse events.

Results Both GTD/EE and DSG/EE groups exhibited minimal effects on the lipid profile. An increased glucose response was noted with both treatments and an increased insulin response was noted with GTD/EE. Hemostatic activity was increased in both groups but a counteracting increase in fibrinolytic activity occurred together with an increase in coagulatory activity. The incidence of adverse events was comparable between groups, and no significant differences in cycle control were observed between groups. No pregnancies occurred in either group.

Conclusions The 24-day GTD 60 μg/EE 15 μg formulation and DSG/EE produced similar effects on hemostatic balance, lipid metabolism and glucose tolerance, and exhibited comparable efficacy and tolerability.

KEY WORDS Lipid profile, Carbohydrate metabolism, Hemostatic balance, Gestodene, Desogestrel, Oral contraceptives

INTRODUCTION

Oral contraceptive (OC) preparations have been associated with an increased risk of thromboembolic disease[1,2], altered lipid metabolism[3] and abnormal glucose tolerance[4]. In an effort to reduce unwanted

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side-effects, the dose of estrogen has been reduced and new progestogens have been introduced. Ethinylestradiol (EE) typically has a beneficial effect on the blood lipid profile, producing increases in high-density lipoprotein cholesterol (HDL-C) levels, and decreases in low-density lipoprotein cholesterol (LDL-C) levels, although the effect of increasing triglycerides is not positive. In addition, EE does not appear to impair glucose tolerance significantly[5]. The newer progestogens, gestodene (GTD) and desogestrel (DSG), inhibit ovulation but possess less androgenic activity than their predecessors[6]. In addition, they are associated with fewer changes in the hemostatic balance, lipid profile and carbohydrate metabolism[7–9]. Low-dose OCs containing GTD 75 μg and EE 30 μg produce multiple moderate-to-small increases in levels of clotting factors VII and X and decreases in levels of anticoagulants such as antithrombin and protein S[9,10]; however, these increases in coagulation measurements appear to be counteracted by an increase in fibrinolytic activity[11].

A new ultra-low-dose combination OC containing GTD 60 μg/EE 15 μg has been developed. When this combination OC is taken in a 24-day regimen, the plasma concentrations of endogenous estradiol, and progesterone are more consistent throughout the cycle than in a 21-day regimen[12]. The primary objective of the present study was to compare the effects of a 24-day regimen of GTD 60 μg/EE 15 μg and a 21-day regimen of DSG 150 μg/EE 20 μg (Mercilon®) on hemostatic balance, blood lipid profile and carbohydrate metabolism over six treatment cycles.

MATERIALS AND METHODS

This phase III, comparative, randomized, parallel-group, open-label study was conducted at multiple centers for six treatment cycles. The study was conducted in accordance with the provisions of the Declaration of Helsinki, and subjects gave informed consent before enrolment.

Subject selection

Women between the ages of 18 and 40 years with regular (25–31 days) menstrual cycles and a body mass index of ≤ 28 kg/m^2 were eligible. Subjects were also required to be either non-smokers or to smoke ten or fewer cigarettes/day. Women with thrombophlebitis, thromboembolic disorders, known or suspected clotting disorders, or hyperlipidemia were excluded from the study. Women with abnormal HDL-C, LDL-C, thyroid hormone, or thyroid-stimulating hormone levels also were excluded.

Study schedule

At the prestudy screening evaluation (visit 1), eligible women provided a medical history and underwent a complete physical examination with routine laboratory determinations and vital signs. Assessment of hemostatic measures, blood lipids and glucose tolerance was also performed between days 17 and 21, inclusive, of the prestudy screening cycle.

At the baseline evaluation (visit 2), a pregnancy test was performed and blood pressure was recorded, and subjects who satisfied inclusion criteria were randomly assigned to receive GTD/EE or DSG/EE. Subjects in the GTD/EE group received a 28-day blister pack containing GTD 60 μg/EE 15 μg pills for 24 days and placebo pills for 4 days. Women in the DSG/EE treatment group received a 21-day blister pack of DSG 150 μg/EE 20 μg pills, which were to be taken on days 1 to 21 of the cycle, to be followed by a 7-day pill-free interval. Each subject began taking study medication on the first day of menses at cycle 1, and subsequent blister packs of study medication were started after day 28 of the previous cycle.

Safety assessment

Hemostatic measures, blood lipid measurements and carbohydrate metabolism were assessed at cycles 3 and 6 within a range of 15 days before to 15 days after the cycle. Angiotensinogen blood concentrations were assessed at cycle 6.

The primary variables for analysis were fibrin degradation products[13] and D-dimer levels (Enzygnost D-dimer, Behringwerke, Marburg, Germany) for hemostatic balance; total cholesterol, HDL-C and subfractions (HDL_2-C, HDL_3-C), LDL-C and triglycerides for blood lipids; and the area under the curve (AUC) for glucose, insulin and C-peptide from the 3-h oral glucose tolerance test. In addition, the following concentrations were analyzed: fibrinogen[14], factor VII total (VIIt)[15], factor VII activated (VIIa)[16], prothrombin fragments (F1 + 2)[17], plasminogen[18],

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plasmin–antiplasmin (PAP) inhibitor complexes[19], a measure of fibrinolytic activity and tissue-type plasminogen activator antigen (tPAag)[20] and plasminogen activator inhibitor 1 antigen (PAI-1ag)[21].

The effects of GTD 60 μg/EE 15 μg and DSG 150 μg/EE 20 μg on hemostatic balance were assessed as described by Kluft[22] and Winkler[23]. Blood collection and handling was done according to the Leiden Fibrinolysis Workshop Criteria[24,25]. Carbohydrate metabolism was evaluated by using a 3-h oral glucose tolerance test[26]. Safety was assessed from adverse events and from changes in routine laboratory determinations and vital signs. Treatment-emergent adverse events were new adverse events, not present at screening, or adverse events present at the prestudy screening that worsened during treatment.

Efficacy and cycle control assessments

The main efficacy assessment was the total number of pregnancies. The pill-pack cycles were analyzed for each treatment cycle from cycles 1 to 6 by using bleeding data reported by subjects on diary cards. Bleeding was classified as normal, breakthrough bleeding, spotting, or absence of bleeding. A cycle was classified as normal if withdrawal bleeding occurred during the 7-day period after the last day of active pill intake (day 21 for DSG/EE and day 24 for GTD/EE) with no breakthrough bleeding or spotting during the rest of the cycle. A withdrawal bleeding episode was defined as a sequence of ≥ 1 day of spotting, bleeding, or both during the 7-day period after the last day of active pill intake, bounded by 2 consecutive days without bleeding. Spotting was defined as very slight bleeding that required no sanitary protection, whereas breakthrough bleeding required sanitary protection.

Data analysis

An intent-to-treat analysis was performed, which included all subjects who were assigned to treatment and who took at least one pill of the assigned study drug. Statistical comparisons of hemostatic measures, blood lipids and glucose tolerance parameters were made by analysis of covariance (ANCOVA) with treatment and investigator as factors and prestudy screening value as a covariate. Within-group changes from screening values were assessed with the paired *t* test. Fisher's exact test was used for comparisons of treatment groups with respect to the incidence of adverse events. Bleeding characteristics were compared between groups for each cycle by the extended Mantel–Haenszel test. All statistical results were presented by use of two-sided tests at the $\alpha = 0.05$ level.

Table 1 Demographic characteristics at the prestudy screening visit

Characteristic	*GTD/EE* (*n* = 62)	*DSG/EE* (*n* = 62)	*p Value*
Age (years)			
Mean ± SD	26.1 ± 4.7	26.3 ± 5.2	0.83
Range	19–40	18–39	
Weight (kg)			
Mean ± SD	63.2 ± 8.4	64.8 ± 7.3	0.28
Range	41.5–80.0	48.0–87.5	
Body mass index (kg/m²)			
Mean ± SD	22.1 ± 2.3	22.5 ± 2.0	0.33
OC status (*n*)			
Recent user*	49 (79%)	41 (66%)	0.26
Former user**	10 (16%)	15 (24%)	
New user	3 (5%)	6 (10%)	
Number of smokers	18	13	0.90
Cigarettes per day			
Mean ± SD	7.3 ± 3.4	7.9 ± 4.9	
Median	10	10	
Range	1–10	1–20	
Alcohol consumption (*n*)			
None	8 (13%)	9 (15%)	0.39
Occasionally	43 (69%)	46 (74%)	
Weekly	10 (16%)	7 (11%)	
Daily	1 (2%)	0	

GTD, gestodene; EE, ethinylestradiol; DSG, desogestrel; OC, oral contraceptive; *OC used within past 3 months; **no OC used in > 3 months

RESULTS

A total of 124 healthy women were randomly assigned to receive GTD 60 μg/EE 15 μg ($n = 62$) or DSG 150 μg/EE 20 μg ($n = 62$). At the prestudy screening visit, the two treatment groups were comparable with respect to age, weight and body mass index (Table 1). A total of 61 women in the GTD/EE group and 59 in the DSG/EE group completed six treatment cycles,

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Table 2 Effect of gestodene/ethinylestradiol (GTD/EE) and desogestrel/ethinylestradiol (DSG/EE) on hemostatic measures

		GTD/EE		*DSG/EE*		
Variable	*Interval*	*Number of subjects*	*Mean ± SD*	*Number of subjects*	*Mean ± SD*	*p Value*[1]
Fibrin degradation products (µg/l)	prestudy	62	135.2 ± 50.9	62	151.5 ± 78.8	ND
	cycle 3	61	189.2 ± 139.4*	61	202.7 ± 120.9**	NS
	cycle 6	59	224.7 ± 284.2*	58	320.4 ± 661.2	NS
D-dimer (µg/l)	prestudy	62	12.3 ± 8.64	62	14.2 ± 11.2	ND
	cycle 3	61	23.8 ± 32.6*	61	24.7 ± 24.9**	NS
	cycle 6	59	28.9 ± 75.6	58	36.9 ± 43.4**	NS
Fibrinogen (g/l)	prestudy	62	2.67 ± 0.55	62	2.73 ± 0.47	ND
	cycle 3	61	3.00 ± 0.58**	61	3.06 ± 0.54**	NS
	cycle 6	59	2.99 ± 0.51**	58	3.00 ± 0.64**	NS
Factor VIIt (%)	prestudy	62	88.3 ± 24.6	62	91.8 ± 20.4	ND
	cycle 3	61	115.2 ± 33.8**	61	120.6 ± 27.0**	NS
	cycle 6	59	119.6 ± 35.8**	58	124.1 ± 28.8**	NS
Factor VIIa (IU/l)	prestudy	62	49.3 ± 28.0	62	53.5 ± 35.1	ND
	cycle 3	61	67.1 ± 44.7**	61	69.4 ± 33.9*	NS
	cycle 6	59	84.4 ± 129.1*	58	90.5 ± 95.3*	NS
F1 + 2 (nmol/l)	prestudy	62	0.86 ± 0.45	62	0.95 ± 0.43	ND
	cycle 3	61	1.11 ± 0.59*	61	1.20 ± 0.49*	NS
	cycle 6	59	1.41 ± 1.05**	58	1.97 ± 2.47*	NS
Plasminogen activity (%)	prestudy	62	113.0 ± 17.4	62	111.5 ± 19.5	ND
	cycle3	61	150.0 ± 22.3**	61	151.6 ± 19.1**	NS
	cycle 6	59	149.8 ± 19.7**	58	152.6 ± 18.0**	NS
PAP (µg/l)	prestudy	62	315.5 ± 96.0	62	354.5 ± 128.4	ND
	cycle3	61	472.3 ± 132.6**	61	495.7 ± 146.5**	NS
	cycle 6	59	506.2 ± 219.2**	58	516.5 ± 280.1**	NS
tPA antigen (µg/l)	prestudy	62	4.91 ± 1.85	62	5.57 ± 1.74	ND
	cycle 3	61	3.10 ± 1.42**	61	3.95 ± 1.94**	0.04
	cycle 6	59	2.86 ± 0.88**	58	3.66 ± 1.63**	0.02
PAI-1 antigen (µg/l)	prestudy	62	42.6 ± 28.1	62	47.4 ± 21.1	ND
	cycle 3	61	17.4 ± 12.9**	61	19.9 ± 14.7**	NS
	cycle 6	59	14.4 ± 7.9**	58	22.2 ± 21.3**	0.02
Angiotensinogen (pmol/l)	prestudy	62	1067.7 ± 242.9	62	1044.3 ± 247.1	ND
	cycle 6	59	2972.2 ± 594.9**	58	3117.2 ± 667.3**	0.07

[1]ANCOVA for between-group differences; *$p < 0.05$; **$p < 0.001$ vs. prestudy values (paired *t* test); ND, not done; NS, not significant

representing a total of 356 cycles for the GTD/EE group and 351 for the DSG/EE group.

Safety

Hemostatic measurements

A statistically significant increase in the levels of fibrin degradation products from the prestudy screening visit to the cycle 3 visit was noted in both groups. At the cycle 6 visit, the difference from prestudy levels was significant in the GTD/EE group but not quite significant in the DSG/EE group ($p = 0.059$) (Table 2). A significant increase in D-dimer levels from prestudy screening to the cycle 3 visit was also noted in both treatment groups. At the cycle 6 visit, the increase in D-dimer levels was significant only in the DSG/EE group. No significant differences were observed between the groups in mean fibrin degradation products and D-dimer values at any visit.

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Table 3 Effect of gestodene/ethinylestradiol (GTD/EE) and desogestrel/ethinylestradiol (DSG/EE) on lipid parameters

		GTD/EE		*DSG/EE*		
Variable	*Interval*	*Number of subjects*	*Mean ± SD*	*Number of subjects*	*Mean ± SD*	*p Value*[†]
Total cholesterol (mmol/l)	prestudy	62	4.66 ± 0.74	62	4.74 ± 0.87	ND
	cycle 3	60	4.47 ± 0.66*	61	4.52 ± 0.68*	NS
	cycle 6	60	4.58 ± 0.75	56	4.56 ± 0.67	NS
LDL cholesterol (mol/l)	prestudy	62	2.66 ± 0.56	62	2.75 ± 0.73	ND
	cycle 3	60	2.45 ± 0.51**	61	2.45 ± 0.61**	NS
	cycle 6	60	2.51 ± 0.61*	56	2.49 ± 0.62*	NS
HDL cholesterol (mmol/l)	prestudy	62	1.58 ± 0.33	62	1.56 ± 0.38	ND
	cycle 3	60	1.57 ± 0.36	61	1.63 ± 0.33*	NS
	cycle 6	60	1.59 ± 0.34	56	1.62 ± 0.31*	NS
HDL_2 cholesterol (mmol/l)	prestudy	62	0.56 ± 0.26	62	0.57 ± 0.31	ND
	cycle 3	60	0.51 ± 0.26	61	0.56 ± 0.26	NS
	cycle 6	60	0.54 ± 0.28	56	0.56 ± 0.23	NS
HDL_3 cholesterol (mmol/l)	prestudy	62	1.02 ± 0.18	62	0.99 ± 0.20	ND
	cycle 3	60	1.06 ± 0.19*	61	1.06 ± 0.20*	NS
	cycle 6	60	1.05 ± 0.17	56	1.06 ± 0.14*	NS
VLDL cholesterol (mmol/l)	prestudy	62	0.34 ± 0.15	62	0.35 ± 0.15	ND
	cycle 3	60	0.35 ± 0.18	61	0.36 ± 0.16	NS
	cycle 6	60	0.40 ± 0.20*	56	0.38 ± 0.21	NS
Triglycerides (mmol/l)	prestudy	62	0.88 ± 0.30	62	0.89 ± 0.32	ND
	cycle 3	60	1.08 ± 0.42**	61	1.15 ± 0.35**	NS
	cycle 6	60	1.15 ± 0.43**	56	1.15 ± 0.41**	NS
Lipoprotein(a) (g/l)	prestudy	62	0.14 ± 0.21	62	0.14 ± 0.15	ND
	cycle 3	60	0.12 ± 0.17	61	0.13 ± 0.15	NS
	cycle 6	60	0.13 ± 0.19	56	0.13 ± 0.16	NS
Apoprotein A1 (g/l)	prestudy	62	1.32 ± 0.18	62	1.29 ± 0.19	ND
	cycle 3	60	1.43 ± 0.20	61	1.51 ± 0.21**	< 0.001
	cycle 6	60	1.47 ± 0.19	56	1.51 ± 0.19**	0.05
Apoprotein B (g/l)	prestudy	62	0.63 ± 0.12	62	0.64 ± 0.15	ND
	cycle 3	60	0.66 ± 0.16	61	0.65 ± 0.15	NS
	cycle 6	60	0.69 ± 0.18	56	0.67 ± 0.15*	NS
LDL cholesterol/apoprotein B	prestudy	62	4.20 ± 0.39	62	4.26 ± 0.37	ND
	cycle 3	60	3.73 ± 0.35	61	3.75 ± 0.28**	NS
	cycle 6	60	3.68 ± 0.32	56	3.72 ± 0.23**	NS
Apoprotein B/apoprotein A1	prestudy	62	0.49 ± 0.11	62	0.51 ± 0.14	ND
	cycle 3	60	0.47 ± 0.14	61	0.44 ± 0.12**	< 0.001
	cycle 6	60	0.47 ± 0.14	56	0.45 ± 0.12**	0.03

[†]ANCOVA for between-group differences; *$p < 0.05$; **$p < 0.001$ vs. prestudy values (paired *t* test); ND, not done; NS, not significant

Mean levels of fibrinogen, factor VIIt, factor VIIa, and F1 + 2 showed significant increases from prestudy screening in both treatment groups at each visit. Significant increases from prestudy screening in mean plasminogen activity and fibrinolytic activity (PAP) values and tPAag were also observed in both treatment groups at the cycle 3 and 6 visits. In addition, there were significant decreases from prestudy screening in mean levels of tPAag and PAI-1ag at both visits.

A significant increase from prestudy in mean angiotensinogen concentration was noted in both groups at cycle 6. The difference was greater in the

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DSG/EE group and the difference between groups approached statistical significance ($p = 0.07$).

Lipid metabolism

For the primary lipid assessments, significant decreases from prestudy screening values were noted in both groups in mean total cholesterol levels at the cycle 3 visit and LDL-C levels at the cycle 3 and 6 visits (Table 3). Significant increases from prestudy screening values were noted in mean triglyceride levels in both treatment groups at both cycle 3 and 6 visits and in HDL-C levels in the DSG/EE group at both visits. The increase in HDL-C levels was mostly accounted for by an increase in HDL_3-C levels. Apoprotein A1 was significantly ($p < 0.001$) increased in both groups at cycles 3 and 6; adjusted means were significantly ($p \leq 0.05$) greater with DSG/EE than with GTD/EE at cycles 3 and 6.

Carbohydrate metabolism

At the cycle 6 visit, the results of the glucose tolerance test showed that the mean glucose AUC increased significantly from prestudy values in both treatment groups, but the mean insulin AUC increased significantly in the GTD/EE group only (Figure 1). There was no significant change from prestudy screening for C-peptide AUC values in either group (Table 4).

Efficacy

No pregnancy was reported in either treatment group after completion of six cycles. The analysis of the bleeding was based on 352 cycles in the GTD/EE group and 341 cycles in the DSG/EE group. Overall, 68% of the cycles were classified as normal for the GTD/EE group compared with 76% of the cycles for the DSG/EE group (Figure 2). No significant difference was noted between the two groups in the incidence of normal cycles. The incidence of breakthrough bleeding and/or spotting, with or without withdrawal bleeding, was 28% in the GTD/EE group compared with 23% in the DSG/EE group. Absence of bleeding was reported in 4% of the cycles for the GTD/EE group and 1% of the cycles for the DSG/EE group.

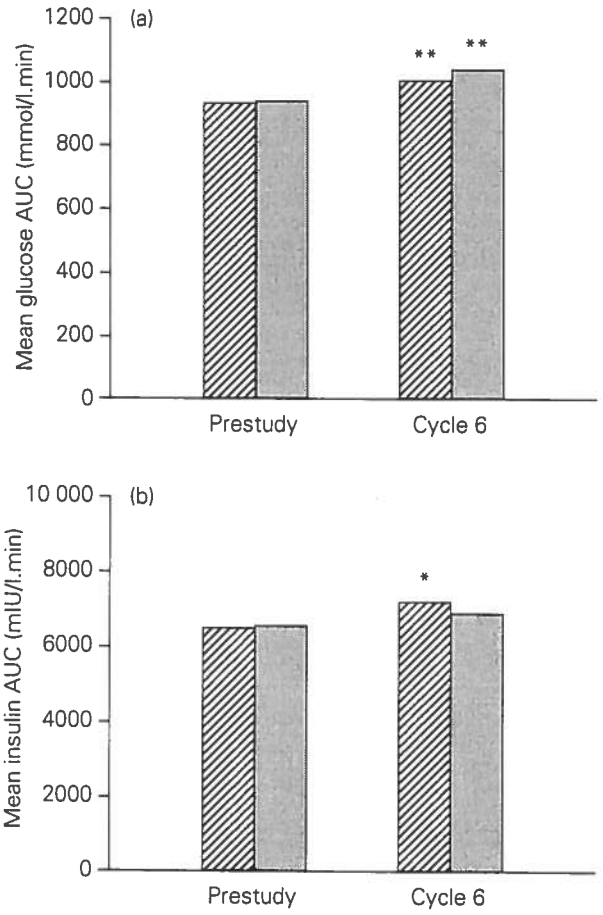

Figure 1 Effects of gestodene/ethinylestradiol (hatched bars) and desogestrel/ethinylestradiol (shaded bars) on fasting glucose (a) and insulin (b) area under the curve (AUC) from prestudy to cycle 6. $*p < 0.05$ vs. prestudy; $**p < 0.001$ vs. prestudy values. Between-group differences were not significant

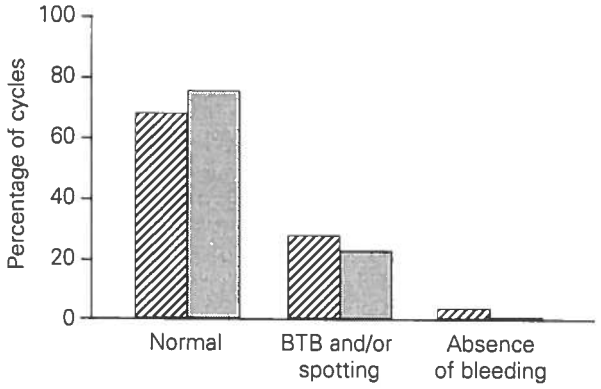

Figure 2 Cyclic bleeding classifications for gestodene/ethinylestradiol (hatched bars; 352 cycles) and desogestrel/ethinylestradiol (shaded bars; 341 cycles) during six cycles. BTB, breakthrough bleeding

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Table 4 Effects of gestodene/ethinylestradiol (GTD/EE) and desogestrel/ethinylestradiol (DSG/EE) on the glucose tolerance test

		GTD/EE		*DSG/EE*		
Variable	*Interval*	*Number of subjects*	*Mean ± SD*	*Number of subjects*	*Mean ± SD*	*p Value*†
Fasting glucose (mmol/l)	prestudy	62	4.86 ± 0.33	62	4.91 ± 0.43	ND
	cycle 6	60	5.01 ± 0.40*	58	5.01 ± 0.42	NS
Fasting insulin (mIU/l)	prestudy	62	8.53 ± 4.67	62	8.90 ± 4.29	ND
	cycle 6	60	10.62 ± 6.03*	58	10.22 ± 4.63*	NS
Fasting C-peptide (pmol/l)	prestudy	62	0.57 ± 0.24	62	0.58 ± 0.20	ND
	cycle 6	60	0.54 ± 0.24	58	0.50 ± 0.22*	NS
Glucose AUC (min.mmol/l)	prestudy	61	942 ± 173	60	947 ± 144	ND
	cycle 6	59	1014 ± 218**	57	1056 ± 153**	NS
Insulin AUC (min.mIU/l)	prestudy	61	6493 ± 4170	60	6599 ± 2643	ND
	cycle 6	59	7309 ± 5035*	57	6982 ± 2833	NS
C-peptide AUC (min.nmol/l)	prestudy	61	303 ± 119	60	325 ± 93.00	ND
	cycle 6	59	300 ± 107	57	299 ± 97.00	NS

†ANOVA for between-group differences, with treatment and investigator as factors at each visit; *$p < 0.05$; **$p < 0.001$ vs. prestudy values; ND, not done; NS, not significant

Tolerability

No serious adverse events possibly related to drug use occurred in either group. At least one treatment-emergent adverse event was reported by 49 (79%) subjects in the GTD/EE group and by 44 (71%) in the DSG/EE group. In both the GTD/EE and DSG/EE groups, the most frequent adverse events were headache (52% and 39%), flu syndrome (23% and 19%), nausea (19% and 16%) and abdominal pain (18% and 15%) (Table 5). No clinically important changes in mean routine laboratory determinations, vital signs or body weight were reported with either group.

Table 5 Most frequent treatment-emergent adverse events with gestodene/ethinylestradiol (GTD/EE) and desogestrel/ethinylestradiol (DSG/EE)

	GTD/EE		*DSG/EE*	
Event	*n*	*%*	*n*	*%*
Headache	32	52	24	39
Flu syndrome	14	23	12	19
Nausea	12	19	10	16
Abdominal pain	11	18	9	15

DISCUSSION

The decrease in adverse effects on blood lipid and carbohydrate profiles is well documented with low-dose oral contraceptive preparations containing newer progestogens[1,27]. The results of the present study showed that both GTD 60 µg/EE 15 µg and DSG 150 µg/EE 20 µg had a counteracting effect on coagulant and fibrinolytic activity. Although the dynamic balance of hemostasis showed a shift from baseline values towards greater activity during treatment in both groups, the increase in coagulatory activity appeared to be counteracted by an increase in fibrinolytic activity. Specifically, the increase in coagulation activity, as indicated by an increase in F1 + 2 levels, was accompanied by an increase in fibrin degradation products and D-dimer levels, suggesting an increase in fibrin turnover and the fibrin deposition/degradation process.

The results presented here also showed that GTD 60 µg/EE 15 µg had a minimal impact on the lipid profile. Total cholesterol and LDL-C levels were unchanged or decreased, while fasting triglyceride levels increased, as is to be expected with administration of an estrogen.

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Importantly, apoprotein A1 levels increased and the ratio of apoproteins B to A1 decreased with GTD/EE and DSG/EE; these changes may be associated with a decreased risk of cardiovascular disease[28]. Overall, the minimal changes in the lipid profile with the GTD 60 µg/EE 15 µg dose were sufficiently small to maintain normal levels of beneficial lipids and would not be expected to adversely affect cardiovascular risk in healthy women[29,30]. The results of the glucose tolerance test showed an increased glucose response with GTD 60 µg/EE 15 µg; however, these changes were minimal with no significant difference between groups. This was consistent with results reported with other marketed gestodene-containing products[31].

The GTD 60 µg/EE 15 µg preparation evaluated in this comparative study combines a 24-day regimen, which shortens the pill-free interval, with the lowest dose of GTD and EE. In comparison with a 21-day regimen, the 24-day regimen of GTD 60 µg/EE 15 µg provides more consistent plasma progesterone and estradiol concentrations throughout the cycle[12]. In this study, the cycle control and safety profile of GTD/EE compared well with those of DSG/EE.

In summary, the 24-day GTD 60 µg/EE 15 µg formulation and the 21-day DSG/EE formulation were comparable in their effects on hemostatic balance, lipid metabolism and glucose tolerance, and the two formulations exhibited comparable efficacy and tolerability. The ultra-low dose combined with a 24-day regimen represents a new approach in contraceptive formulations and provides a new option for women requesting oral contraception. This formulation has the potential for improved tolerability, which may be beneficial for patient compliance, while providing an ultra-low steroid dose to reduce the potential for OC complications.

REFERENCES

1. Speroff L, DeCherney A. Evaluation of a new generation of oral contraceptives. *Obstet Gynecol* 1993; 81: 1034–47
2. Daly L, Bonnar J. Comparative studies of 30 micrograms ethinylestradiol combined with gestodene and desogestrel on blood coagulation, fibrinolysis, and platelets. *Am J Obstet Gynecol* 1990; 163: 430–7
3. Fotherby K, Caldwell A. New progestogens in oral contraception. *Contraception* 1994; 49: 1–32
4. Wynn V, Doar J. Some effects of oral contraceptives on carbohydrate metabolism. *Lancet* 1966; 2: 715–19
5. Spellacy W, Buhi W, Birk S. Effect of estrogen treatment for one year on carbohydrate and lipid metabolism in women with normal and abnormal glucose tolerance test results. *Am J Obstet Gynecol* 1978; 131: 87–90
6. Rebar R, Zeserson K. Characteristics of the new progestogens in combination oral contraceptive. *Contraception* 1991; 44: 1–10
7. Godsland I, Crook D, Simpson R, *et al.* The effects of different formulations of oral contraceptive agents on lipid and carbohydrate metabolism. *N Engl J Med* 1990; 323: 1375–81
8. Kaplan B. Desogestrel, norgestimate, and gestodene: the newer progestins. *Ann Pharmacother* 1995; 29: 736–42
9. Kluft C, Lansink M. Effect of oral contraceptives on haemostasis variables. *Thromb Haemost* 1997; 78: 315–26
10. Shoupe D. New progestins – clinical experiences: gestodene. *Am J Obstet Gynecol* 1994; 170: 1562–8
11. Fotherby K. Desogestrel and gestodene in oral contraception: a review of European experience. *J Drug Dev* 1991; 4: 101–11
12. Sullivan H, Furniss H, Spona J, Elstein M. Effect of 21-day and 24-day oral contraceptive regimens containing gestodene (60 µg) and ethinyl estradiol (15 µg) on ovarian activity. *Fertil Steril* 1999; 72: 115–20
13. Koppert PW, Hoegee-de Nobel E, Nieuwenhuizen W. A monoclonal antibody-based enzyme immunoassay for fibrin degradation products in plasma. *Thromb Haemost* 1988; 59: 310–15
14. Clauss A von. Gerinnungsphysiologische Schnellmethode zur Bestimmung des Fibrinogens. *Acta Haematol* 1957; 17: 237–46
15. Avvisati G. Evaluation of a new chromogenic assay for factor VII and its application in patients on oral anticoagulant treatment. *Br J Haematol* 1980; 45: 343–52
16. Morrissey JH, Macik BG, Neuenschwander PF, *et al.* Quantitation of activated factor VII levels in plasma using a tissue factor mutant selectively deficient in promoting factor VII activation. *Blood* 1993; 81: 734–44

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17. Pelzer H, Schwarz A, Stuber W. Determination of human prothrombin fragment in plasma with an antibody against a synthetic peptide. *Thromb Haemost* 1991; 65: 153–9
18. Gram J, Jespersen J. A functional plasminogen assay utilising the potentiating effect of fibrinogen to correct for the overestimation of plasminogen in pathological plasma samples. *Thromb Haemost* 1985; 53: 255–9
19. Pelzer H, Pilgrim A, Schwarz A, *et al.* Determination of alpha-2-antiplasmin complex in human plasma with an enzyme-linked immunosorbent assay. *Fibrinolysis* 1993; 7: 69–74
20. Ranby M, Nguyen C, Scarabin PY, *et al.* Immunoreactivity of tissue plasminogen activator and of its inhibitor complexes. Biochemical and multicenter validation of a two site immunosorbent assay. *Thromb Haemost* 1989; 61: 409–14
21. Meijer P, Pollet DE, Wauters J, *et al.* Specificity of antigen assays of plasminogen activator inhibitor in plasma: innotest PAI-1 immunoassay evaluated. *Clin Chem* 1994; 40: 110–15
22. Kluft C. Disorders of the hemostatic system and the risk of the development of thrombotic and cardiovascular diseases: limitations of laboratory diagnosis. *Am J Obstet Gynecol* 1990; 163: 305–12
23. Winkler UH, Koslowski S, Oberhoff C, *et al.* Changes of the dynamic equilibrium of hemostasis associated with the use of low-dose oral contraceptives. *Adv Contracept* 1991; 7(Suppl 3): 273–84
24. Kluft C, Verheijen JH. Leiden fibrinolysis working party: blood collection and handling procedures for assessment of tissue-type plasminogen activator (t-PA) and plasminogen activator inhibitor (PAI 1). *Fibrinolysis* 1990; 4: 155–61
25. Kluft C, Meijer P. Update 1996: blood collection and handling procedures for assessment of plasminogen activators and and inhibitors (Leiden Fibrinolysis Workshop). *Fibrinolysis* 1996; 10 (Suppl): 2171–9
26. Petersen KR, Sidelmann J, Skouby SO, *et al.* Effects of monophasic low-dose oral contraceptives on fibrin formation and resolution in young women. *Am J Obstet Gynecol* 1993; 168(part 1): 32–8
27. Baird D, Glasier A. Hormonal contraception. *N Engl J Med* 1993; 328: 1543–9
28. Brown G, Albers JJ, Fisher LD, *et al.* Regression of coronary artery disease as a result of intensive lipid-lowering therapy in men with high levels of apolipoprotein B. *N Engl J Med* 1990; 323: 1289–95
29. Walsh J, Grady D. Treatment of hyperlipidemia in women. *J Am Med Assoc* 1995; 274: 1152–8
30. Marsh MS, Crook D, Whitcroft SIJ, *et al.* Effect of continuous combined estrogen and desogestrel hormone replacement therapy on serum lipids and lipoproteins. *Obstet Gynecol* 1994; 83: 19–23
31. Spellacy W, Tsibris A, Tsibris J, *et al.* Carbohydrate metabolism studies after one year of using an oral contraceptive containing gestodene and ethinylestradiol. *Contraception* 1994; 49: 125–30

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